

Many a time, it's necessary to steer an input pin to a known state if no input is present. This can be done by adding a pull-up resistor (to +5V) or a pull-down resistor (resistor to ground) on the input. A 10k resistor is a good choice as a pull-up or pull-down resistor.

Code:

```
• int ledPin = 13;
void setup() {
    pinMode(ledPin, OUTPUT);    // sets the digital pin 13
    as output
}
void loop() {
    digitalWrite(ledPin, HIGH); // write logic HIGH to pin 13,
    making output voltage +5 V
}
```

Digital Read

To read data using the digital pins, use the following syntax.

- `digitalRead (pin)`

This reads the value from the specified digital pin, either HIGH or LOW.

Code:

```
• int inPin = 7;    // pushbutton connected to digital pin 7
int val = 0;        // variable to store the read value
void setup() {
    pinMode(inPin, INPUT);    // sets the digital pin 7 as
    input
}
void loop() {
    val = digitalRead(inPin); // read the value at input pin
    and store in variable val
}
```

Analog Pins

Analog pins on the Arduino™ only read analog data. They have an Analog-to-Digital converter connected to them, which reads the voltage at the pin and converts it to digital form. Since it's a 10-bit Digital-to-Analog converter, the resulting range of 'analog read' function is 0 to 1023.

The syntax to use analog read is:

- `analogRead(pin);`